

MERIDIAN DATA CENTRE — PHASE 1

PROJECT SPECIFICATION (SPEC-MDC-P1-RevC)

Discipline: Multidiscipline (Div 26 / 23 / 28 / 27 / 03)

Sheet: Construction-Issue Authorisation

Distribution: EPC Contractor, Owner, Commissioning Agent



Reviewed / Approved:

Lead Electrical Engineer — A. Rao

Signature / Date

Lead Mechanical Engineer — P. Nair

Signature / Date

Project Director — M. Fernandes

Signature / Date

Commissioning Authority (CxA) — K. Iyer

Signature / Date

This sheet authorises the accompanying specification for construction issue.

MERIDIAN DATA CENTRE

Phase 1 — Nashik (24 MW, Rated-3 / Tier III)

PROJECT SPECIFICATION — TECHNICAL REQUIREMENTS

Document No.	SPEC-MDC-P1-RevC	Revision	C — For Construction
Discipline	Multidiscipline (26/23/28/27/03)	Date	2026-05-30
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This specification establishes the minimum technical requirements for the Works. Where a requirement states a limiting value (e.g. 'not less than', 'shall not exceed'), that value is a compliance threshold. Vendor submittals are assessed for conformity against the clauses herein. Refer to the preceding stamped sheet for construction-issue authorisation.

DIVISION 26 — ELECTRICAL SYSTEMS

Section 26 33 53 — Static Uninterruptible Power Supply (UPS)

ELEC-4.2.1 UPS — Rated Output Capacity

Each static UPS module (tag UPS-02A) **shall** provide a rated output apparent power of **not less than 2.0 MVA** at an output power factor of 0.9, referenced to nominal conditions of 400 V, 50 Hz and 25 °C. No negative tolerance on rated capacity is permitted. Capacity is a critical parameter as the module directly supports the Rated-3 (TIA-942) / Tier III (Uptime) critical bus.

Criticality: CRITICAL. Mandatory (Contractor shall comply).

ELEC-4.2.2 UPS — Inverter-to-Bypass Transfer Time

The synchronised transfer time from inverter to static bypass **shall not exceed 4 milliseconds**. Transfers exceeding this limit are not acceptable as they risk dropout of sensitive IT load. Tolerance: 0 ms above the stated maximum.

Criticality: CRITICAL. Mandatory (Contractor shall comply).

ELEC-4.2.3 UPS — Input Current Harmonic Distortion

Input current total harmonic distortion (iTHD) reflected to the source **shall not exceed 3 %** at 100 % rated linear load, with the input filter in service. Values above 3 % require corrective filtering at the Contractor's cost.

Criticality: MAJOR. Mandatory (Contractor shall comply).

ELEC-4.2.4 UPS — Double-Conversion Efficiency

On-line double-conversion efficiency **shall be not less than 96 %** at 100 % rated load and nominal input voltage. Economy/eco-mode efficiency shall be not less than 98.5 %.

Criticality: MAJOR. Mandatory (Contractor shall comply).

ELEC-4.2.5 UPS — Battery Autonomy

Stored-energy battery autonomy **shall be not less than 5 minutes** at 100 % rated load to the inverter shutdown threshold, sized at the specified end-of-life and design temperature. Autonomy is measured from mains failure to end of discharge.

Criticality: MAJOR. Mandatory (Contractor shall comply).

ELEC-4.2.6 UPS — Output Voltage & Regulation

Nominal output voltage **shall be 400 V** three-phase, four-wire, with steady-state regulation within ± 1 %. Output voltage THD with 100 % linear load shall not exceed 2 %.

Criticality: MINOR. Mandatory (Contractor shall comply).

ELEC-4.2.7 UPS — Overload Withstand

The UPS **shall** sustain an overload of **at least 125 % of rated load for 10 minutes** on the inverter before transfer to bypass, and 150 % for 1 minute.

Criticality: MINOR. Mandatory (Contractor shall comply).

ELEC-4.2.8 UPS — Redundancy & Parallel Operation

The UPS system **shall** be configured for N+1 redundancy and be capable of parallel operation of not less than 4 modules on a common output bus with automatic load sharing.

Criticality: MAJOR. Mandatory (Contractor shall comply).

Section 26 32 13 — Engine Generators

ELEC-5.1.1 Standby Generator — Rated Capacity

Each standby diesel generator set **shall** have a prime/standby rating of **not less than 2500 kVA** at 0.8 power factor, 415 V, 50 Hz, at site reference conditions, sized to carry the full designed critical and mechanical block load.

Criticality: CRITICAL. Mandatory (Contractor shall comply).

ELEC-5.1.2 Standby Generator — Start & Load Acceptance

The generator **shall** reach rated voltage and frequency and be ready to accept load within **10 seconds** of a start signal, and accept the first load step per ISO 8528-5 class G3.

Criticality: MAJOR. Mandatory (Contractor shall comply).

ELEC-5.1.3 Standby Generator — Fuel Autonomy

On-site fuel storage **shall** provide **not less than 48 hours** of generator operation at 100 % rated standby load without replenishment.

Criticality: MAJOR. Mandatory (Contractor shall comply).

Section 26 13 00 — Medium-Voltage Switchgear

ELEC-6.1.1 MV Switchgear — Rated Voltage & Short-Circuit

Medium-voltage switchgear **shall** be rated for a nominal system voltage of 11 kV and a short-circuit withstand of **not less than 25 kA** for 3 seconds.

Criticality: MAJOR. Mandatory (Contractor shall comply).

ELEC-6.1.2 MV Switchgear — Arc-Flash Containment

Switchgear **shall** be internal-arc classified to IEC 62271-200 class IAC AFLR, and arc-flash incident-energy labelling to IEEE 1584 **shall** be provided prior to energisation.

Criticality: MAJOR. Mandatory (Contractor shall comply).

ELEC-6.2.1 Power Distribution — Voltage Drop

Steady-state voltage drop from the LV board to the furthest critical load **shall not exceed 3 %** at full load.

Criticality: MINOR. Mandatory (Contractor shall comply).

ELEC-6.3.1 Earthing & Bonding — Earth Resistance

The site earthing system **shall** achieve a measured earth electrode resistance of **not more than 1 ohm**, verified by fall-of-potential test to IEEE 81.

Criticality: MAJOR. Mandatory (Contractor shall comply).

DIVISION 23 — MECHANICAL & COOLING SYSTEMS

Section 23 64 00 — Water Chillers

MECH-3.1.1 Chiller — Rated Cooling Capacity

Each water-cooled chiller **shall** provide a nominal cooling capacity of **not less than 1500 kW** at the specified chilled-water design conditions (supply 18 °C / return 30 °C).

Criticality: CRITICAL. Mandatory (Contractor shall comply).

MECH-3.1.2 Chiller — Efficiency (COP)

Chiller full-load coefficient of performance (COP) **shall be not less than 5.8** at rated conditions; integrated part-load value (IPLV) not less than 7.0.

Criticality: MAJOR. Mandatory (Contractor shall comply).

MECH-3.2.1 Cooling — System Redundancy

The cooling plant **shall** be configured to N+1 redundancy such that failure of any one chiller does not reduce delivered capacity below 100 % of the design critical load.

Criticality: CRITICAL. Mandatory (Contractor shall comply).

MECH-4.1.1 CRAH Units — Airflow & Sensible Capacity

Each computer-room air handler (CRAH) **shall** deliver a nominal airflow of **not less than 30000 m3/h** with a sensible cooling capacity not less than 150 kW at 24 °C return.

Criticality: MAJOR. Mandatory (Contractor shall comply).

MECH-4.2.1 Chilled Water — Concurrent Maintainability

Chilled-water distribution **shall** be dual-path and valved to permit concurrent maintenance of any pipe section, pump or chiller without interruption to the critical load, consistent with Rated-3 / Tier III.

Criticality: CRITICAL. Mandatory (Contractor shall comply).

MECH-4.3.1 Data Hall — Environmental Envelope

The data hall **shall** be maintained within the ASHRAE recommended envelope: dry-bulb 18 °C to 27 °C and relative humidity within 20 % to 60 %, measured at the equipment inlet.

Criticality: MAJOR. Mandatory (Contractor shall comply).

DIVISION 28 & 21 — FIRE DETECTION & SUPPRESSION

FIRE-2.1.1 Detection — Very-Early Smoke Detection

Data halls and electrical rooms **shall** be provided with aspirating very-early smoke detection (VESDA) with a sensitivity of **not less than 0.005 %/m** obscuration at the most sensitive setting.

Criticality: MAJOR. Mandatory (Contractor shall comply).

FIRE-3.1.1 Suppression — Clean-Agent Concentration

Clean-agent gaseous suppression **shall** achieve the design extinguishing concentration within **10 seconds** of release and maintain a hold time of not less than 10 minutes.

Criticality: CRITICAL. Mandatory (Contractor shall comply).

FIRE-3.2.1 Suppression — Agent Design Concentration

The clean-agent design concentration **shall be not less than 7.0 %** by volume for the protected data-hall risk, with a safety factor per ISO 14520.

Criticality: MAJOR. Mandatory (Contractor shall comply).

FIRE-4.1.1 Fire Rating — Compartmentation

Fire-rated separation between data halls and support spaces **shall** provide a fire resistance of **not less than 120 minutes**.

Criticality: MAJOR. Mandatory (Contractor shall comply).

DIVISION 27 — IT & TELECOMMUNICATIONS INFRASTRUCTURE

IT-2.1.1 Structured Cabling — Backbone Performance

Backbone optical cabling **shall** be OM4 multimode / OS2 single-mode supporting a channel bandwidth of **not less than 100 Gbps** over the designed distances.

Criticality: MINOR. Mandatory (Contractor shall comply).

IT-2.2.1 Pathways — Redundant Diverse Routing

Telecommunications pathways **shall** provide two physically diverse routes between spaces, with a minimum separation of **not less than 20 metres** where practicable, to meet Rated-3 / Tier III path redundancy.

Criticality: MAJOR. Mandatory (Contractor shall comply).

IT-3.1.1 Rack Power — Design Density

The white-space design **shall** support an average rack power density of **not less than 8 kW** per rack with provision for high-density zones to 20 kW per rack.

Criticality: MINOR. Mandatory (Contractor shall comply).

DIVISION 03 & 13 — CIVIL & STRUCTURAL

CIVIL-2.1.1 Data Hall — Floor Live Loading

The data-hall structural floor **shall** be designed for a uniformly distributed live load of **not less than 12 kN/m²**, with a concentrated load rating suitable for loaded IT cabinets.

Criticality: MAJOR. Mandatory (Contractor shall comply).

CIVIL-2.2.1 Raised Access Floor — Loading

Raised access flooring **shall** have a concentrated load rating of **not less than 5 kN** per panel with a safety factor of 2.0.

Criticality: MINOR. Mandatory (Contractor shall comply).

CIVIL-3.1.1 Seismic — Design Category

Equipment anchorage and structural design **shall** comply with the applicable seismic provisions for the site, with an importance factor of **not less than 1.5** for critical facility category.

Criticality: MAJOR. Mandatory (Contractor shall comply).

CIVIL-3.2.1 Building — Water Ingress Protection

The building envelope and critical rooms **shall** be designed to prevent water ingress, with critical electrical rooms located above the design flood level by **not less than 300 mm**.

Criticality: MINOR. Mandatory (Contractor shall comply).

END OF SPECIFICATION — SPEC-MDC-P1-RevC. Limiting values stated in the clauses above constitute mandatory compliance thresholds for vendor submittal review.